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Department of Applied Chemistry (2024-25)

NOTES

UNIT-II

Electrochemistry

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Department of Applied Chemistry
2024-25

Unit II: Electrochemistry

Syllabus

Introduction, Redox reactions, EMF of a cell, standard electrode potential, Nernst equation, numerical and applications to chemical cells. Conductance in electrolytic solutions, specific and molar conductivity, variations of conductivity with concentration, Electrolysis, laws of electrolysis and numerical.
Industrial applications: Electroplating, Electrolytic refining.
Corrosion: Definition, Causes, theories of corrosion- dry, wet and differential aeration.
Contemporary issues related to the topic.

Introduction

Electrochemistry is the study of production of electricity from energy released during spontaneous chemical reactions and the use of electrical energy to bring about non-spontaneous chemical transformations. The subject is of importance both for theoretical and practical considerations.

Applications of Electrochemistry

- Metallurgy.
- Electroplating.
- Batteries.
- Fuel cells.
- Analytical chemistry.
- Biological research.

Metallic and Electrolytic Conduction

Sr .No.	Metallic conduction	Electrolytic conduction
1.	The flow of electricity takes place without decomposition of the substance.	The flow of electricity takes place with decomposition of the substance.
2.	There is no actual transport of matter.	Transfer of matter takes place in the form of ions.
3.	The conduction of electricity is due to the flow of electrons.	The conduction of electricity is due to the flow of ions in the solution.
4.	Metallic conductivity decreases with increase in temperature.	Electrolytic conduction increases with increases in temperature.
5.	Metallic conduction is a physical process that means no new substance is formed.	Electrolytic conduction is both physical and chemical change.

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Electrode

Electrodes can be defined as conductors that are used to make electrical contact with a non-metallic part of the circuit, an electrode is a point where the current enters and leaves the electrolyte.

Types and Examples of Electrodes

There are mainly two types of electrodes, namely reactive and inert electrodes.

- A **reactive electrode** is an electrode which actively participates in the reaction. Some reactive electrodes include zinc, copper, lead and silver.
- An **inert electrode** is a type of electrode which does not participate in a chemical reaction. Some commonly used inert electrodes include platinum, gold, graphite(carbon), and rhodium.

Electrode potential

When a pure metal is dipped in a salt containing its own ions (eg.Zn/ Zn^{2+}) the metal may show any one type of the following tendency (in presence of an external driving force).

$M \rightarrow M^{n+} + ne^-$ (De-electronation, more tendency to undergo oxidation)

$M^{n+} + ne^- = M$ (Electronation, more tendency to resist oxidation & undergo reduction)

When such a reaction takes place equilibrium is established after some time and an electrical double layer is formed.

Certain metals show electronation tendency while some have de-electronation tendency and in both the above cases a potential is set up which is called as "electrode Potential". It is exactly defined as the measure of the tendency of an electrode to lose or gain electrons, when it is in contact with a solution of its own ions of unit concentration at 25^0C. Consequently, the tendency of an electrode to gain electrons indicates reduction (electronation) and such an electrode is termed as 'cathode' the tendency of an electrode to lose electrons indicates reduction (de-electronation) and such an electrode is termed as'anode'. For measurement of electrode potential of an electrode at 25 ^{0} c and unit molar conc. of its own salt, it is connected to std. Hydrogen electrode whose electrode potential is arbitrarily taken as zero. The standard reduction potentials so obtained are listed in order of increasing reduction potential and the series so obtained is called as the Standard Electrochemical or E.M.F. series.

Nernst equation

The Nernst equation provides the relation between the cell potential of an electrochemical cell, the standard cell potential, temperature, and the reaction quotient.

It is given by $-E_{cell} = E^0 - [RT/nF] \ln Q$

$E_{cell} = E^0 - [RT/nF] \ln [P]/[R]$

$E_{cell} = E^0 - [2.303RT/nF] \log[P]/[R]$

Where,

- E_{cell} = cell potential
- E^0 = cell potential under standard conditions
- R = universal gas constant
- T = temperature
- n = number of electrons transferred in the redox reaction
- F = 96487 (≈96500) coulomb/mole = charged carried by one mole of electrons.
- Q = reaction quotient

Consider the following reduction reaction

$M^{n+} + ne^- \rightarrow M(s)$

The potential developed on this electrode is;

$E_{red} = E_{Mn+/M} = E^o_{Mn+/M} - [2.303RT/nF] \log [M]/[M^{n+}]$

Where, $[M^{n+}]$ = active mass of the ions. For simplicity, it may be taken as equal to the molar concentration of the salt.

$E_{red} = E_{Mn+/M} = E^o_{Mn+/M} - [2.303RT/nF] \log 1/[M^{n+}]$……**(1)** {Since [M] = 1}

At 25^oC, T = 25+273 =298 K

Since, R = 8.314 J/K Mol and F = 96500

Therefore, 2.303 RT/F = 0.0591 putting in eq. (1)

$E_{red} = E_{Mn+/M} = E^o_{Mn+/M} - [0.0591/n] \log 1/[M^{n+}]$

E(cell) = e.m.f of cell
E(right) = reduction potential of right hand side electrode (cathode)
E(left) = reduction potential of left hand side electrode (anode)

$E_{cell} = E_{ox}$ (anode) + E_{Red} (cathode)
$E_{cell} = E_R - E_L$ (both are reduction potentials)
$E_{cell} = E_L$ (anode) - E_R (cathode) (both are oxidation potentials)

Therefore, EMF of the cell of a battery depend on 1) The potential difference between the cathode and anode – the EMF of the battery is higher if the the potential difference between the cathode and anode are more. 2) The ratio of the ionic concentration of [M] & [Mn+]- the EMF of the battery decreases with the increasing molar concentration of [M] 3) The temperature- Emf of the cells gets reduced marginally with increase in temperature

Applications of Nernst equation:

1. It can be used to study the effect of electrolyte concentration on electrode potential.$E = E^o - RT/nF \ln[1/M^{n+}]$
2. It can also be used for the calculation of the potential of a cell under non-standardconditions.
3. Determination of unknown concentration of one of the ionic species in a cell is possible with the help of Nernst equation, provided E_{cell} and concentration of other ionic species are known.
4. The pH of a solution can be calculated from the measurement of EMF and Nernstequation.
5. Nernst equation can also used for finding the valence of an ion or the number of electronsinvolved in the electrode reaction.

Conductance in electrolytic solutions

The conductance of electricity by ions present in the solutions is called electrolytic or ionic conductance.

The conductivity of electrolytic solutions depends on:

(i) The nature of the electrolyte added

(ii) Size of the ions produced and their solvation

(iii) The nature of the solvent and its viscosity

(iv) Concentration of the electrolyte

(v) Temperature (it increases with the increase of temperature)

Specific and Molar Conductivity

Specific conductivity or conductivity of an electrolytic solution at any given concentration is the conductance of unit volume of solution kept between two platinum electrodes with a unit area of cross-section and at a distance of unit length.

It is denoted by kappa(K). SI unit of specific conductivity is Sm^{-1} or Scm^{-1}.

Molar conductivity of a solution at a given concentration is the conductance of volume V of a solution containing 1 mole of the electrolyte, kept between two electrodes with the cross-sectional area of A and distance of unit length.

It is denoted as Λ_m. The SI unit of molar conductivity is Sm^2mol^{-1}.

$$\Lambda_m = K/c$$

Here, c = concentration in moles per volume.

κ = specific conductivity and Λ_m = molar conductivity.
As the solution contains only one mole of electrolyte, $\Lambda_m = \kappa V$

Variations of conductivity with concentration

Specific conductivity decreases with a decrease in concentration or on dilution as the number of ions per unit volume that carry the current in a solution decreases.

Molar conductivity increases with a decrease in concentration or on dilution as the total volume of the solution containing one mole of electrolyte increases.

Electrolysis

It is the process by which electric current is passed through a substance to effect a chemical change, in which the substance loses or gains an electron (oxidation or reduction). The process is carried out in an electrolytic cell, an apparatus consisting of positive and negative electrodes held apart and dipped into a solution containing positively and negatively charged ions.

Faradays First Law of Electrolysis

The amount of substance liberated or deposited at each electrode during electrolysis is directly proportions to the quantity of electricity i.e. charge passed through the electrolyte.

$$W \alpha Q \qquad W = Z Q$$

Where, Q is quantity of charge passed through the solution measured in coulomb

and Z is proportionality constant known as electrochemical equivalence(ECE).

Q is the product of current in Amp and time in second. Therefore,

$$Q = I \times t \text{ and } W = Z I t.$$

Electrochemical equivalent can be defined as the amount of substance deposited by passing 1 coulomb of electricity.

Faradays 1st law is independent of temperature, pressure and nature of non-interfering solvent or electrode.

Faradays Second Law of Electrolysis

When the same amount of electricity is passed through different cells containing different electrolytes and arranged in series the amount of substance liberated or deposited at the respective electrodes are directly proportional to their chemical equivalent weights.

$$\frac{W1}{W2} = \frac{E1}{E2}$$

Where, W_1= Weight of substance deposited

W_2= weight of substance liberated
E_1 = Chemical equivalent weight of substance deposited
E_1 = Chemical equivalent weight of substance liberated

Conclusion

1) Because one Faraday (96,500 coulombs) deposits one gram equivalent of the substance, the electrochemical equivalent can be calculated from the equivalent weight i.e.,

$$Z \text{ (electrochemical equivalent)} = \frac{Equivalent\ weight\ of\ the\ substance}{96500}$$

2) Knowing the weight of the substance deposited (W gram) when a definite quantity of electricity (Q coulombs) is passed, the equivalent weight of the substance can be calculated, i.e,

$$\text{Equivalent weight} = \frac{W}{Q} \times 96500$$

Industrial applications of Electrolysis

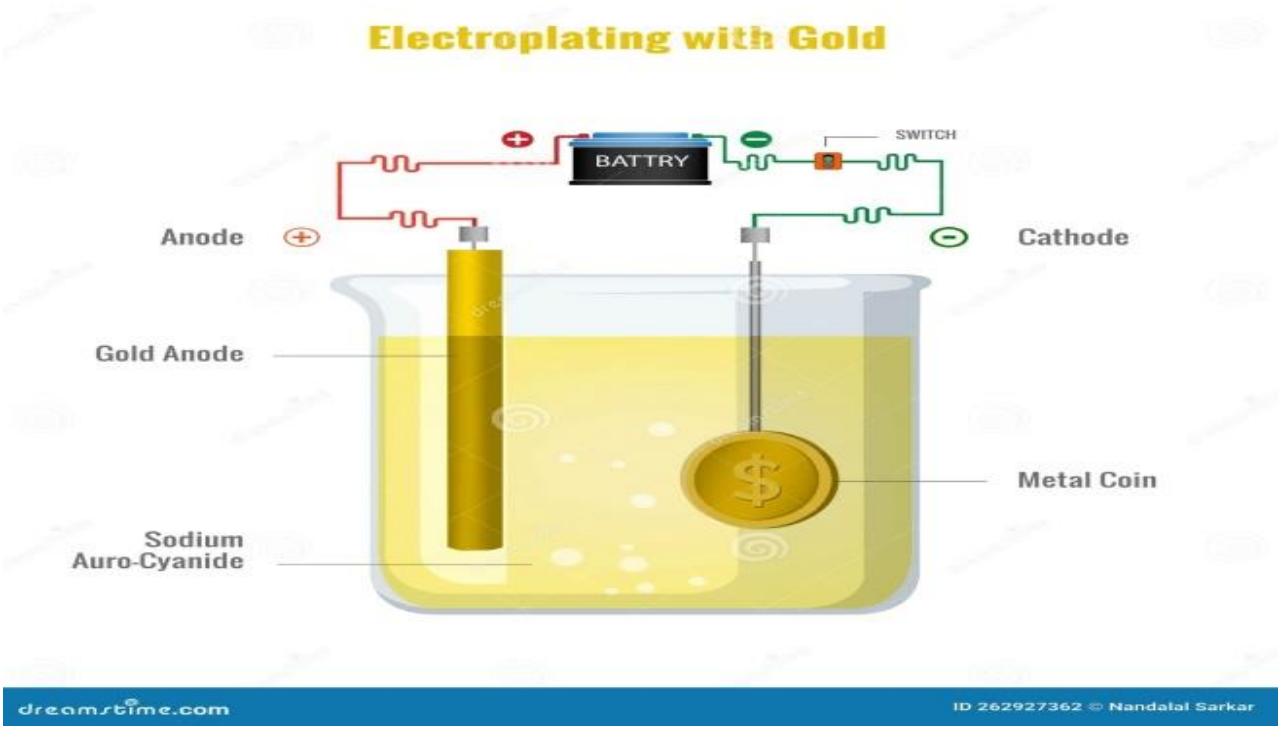
i) Electroplating, ii) Electrolytic refining.

i) Electroplating

- Electroplating is the process of plating a metal onto the other by hydrolysis mostly to prevent corrosion of metal or for decorative purposes.
- The process uses an electric current to reduce dissolved metal cations to develop a lean coherent metal coating on the electrode.
- Electroplating is often applied in the electrical oxidation of anions on a solid substrate like the formation of silver chloride on silver wire to form silver chloride electrodes.
- Electroplating is majorly applied to modify the surface features of an object (e.g corrosion protection, lubricity, abrasion) but the process can also be used to build thickness or make objects by electro forming.
- The electroplating process uses an anode and a cathode. In electroplating, the metal dissolved from the anode can be plated onto the cathode.
- The anode is provided with direct current, oxidizing and dissolving its metal atoms in the electrolyte solution. At the cathode, the dissolved metal ions are reduced and the metal is placed on the product.
- Example of a gold coating- a layer of gold is to be electrodeposited on metallic jwellery to enhance its appearance. Usually, the gold plating is connected to the anode (+ve charged electrode) of the circuit and the jewellery is kept at the cathode (-ve charged electrode).
- Both are kept immersed in a highly developed electrolytic bath (solution). At this stage, a DC current is supplied to the anode that oxidizes the gold atoms and dissolves them into the solution.

- The dissolved ions of gold are reduced at the cathode and plated on the jewellery. However, there are major factors that influence the plating. These includes:

1. The voltage level of current.
2. The temperature and chemical composition of the bath.
3. The current length of time.
4. The distance between the cathode and the anode.



Advantages of Electroplating

Talking about the uses of electroplating, apart from enhancing the appearance of the substrate it is used in various other purposes as well. The major application is to optimize a material's resistance towards corrosion. The plated layer often serves as a sacrificial coating which reveals that it dissolves before the base substance.

Some of the other common applications of electroplating involve:

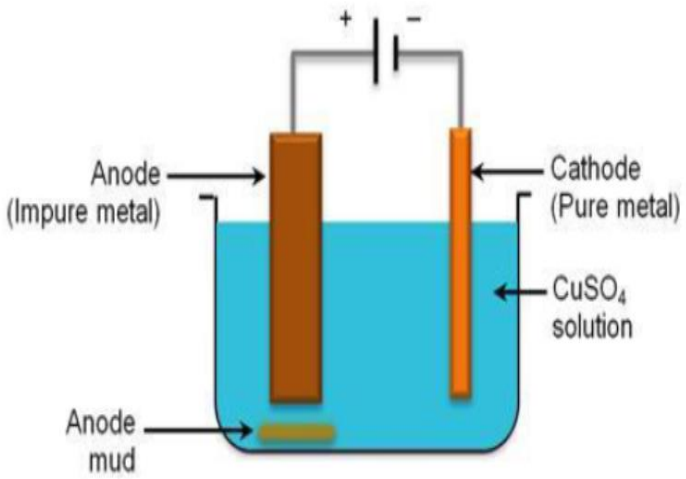
1. Improving wear resistance.
2. Improving the thickness of the metal surface.
3. Enhancing the electrical conductivity like plating a copper layer on an electrical component.
4. Minimizing Friction.
5. Improving surface uniformity.

Industrial applications of electroplating

1. Medical industry- To enhance component biocompabilty Ti, Au, Ag metals are used . For example-. Joint replacement and implants
2. Automotive industry-To protect from corrosion for eg. Zn-Ni plating, Nickel plating with plastic parts
3. Aerospace Industry- for eg. Titanium is used due to appropriate strength to weight ratio, Copper is used for heat resistance, Nickel is used for protection against wear and corrosion.
4. Oil and Gas industry- Ni plating on pipelines
5. Electronics Industry- Gold plating – Connectors & semiconductors to ensure that the item is conductive. Cu plating & palladium alloy plating are commonly used in electronic gadgets.

ii) Electrolytic refining

- Electrolytic refining is a technique used for the extraction and purification of metals by the process of electrolysis. Metals like copper, nickel, gold, lead, silver, and zinc can be purified using electrolytic refining.



Electrolytic refining of Copper

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- In electrolytic refining, the anode is the impure metal to be purified, while the cathode is a thin strip of high purity metal. The electrolyte used in the solution is metal salt. When electric

current passes through the electrolyte, pure metal from the anode dissolves in the electrolyte and gets deposited at the cathode.

- The insoluble impurities settle down at the bottom of the anode and are known as anode mud. The soluble impurities go back into the solution.
- The electrolyte and other conditions must be appropriately selected so that anodic dissolution and deposition of the metal occurs with high efficiency, without impure metals getting transferred from the cathode to the anode.
- Wherever necessary, additives should be added to the electrolyte to enforce correct behaviour at both electrodes. Chloride ion is a common additive to enhance dissolution.
- Impure copper is used as the anode, while a thin strip of highly pure copper is used as the cathode. Copper sulphate solution is used as the electrolyte. When electric current passes through the electrolyte, pure copper from the anode dissolves into the electrolyte, and the same amount of copper gets deposited at the cathode.
- Thus, the thickness of the cathode increases with time. The soluble impurities go into the solution, and the insoluble impurities settle down as anode mud. The anode mud contains antimony, selenium, tellurium, silver, gold, and platinum.

- **Reaction occurring at the cathode:** $Cu^{2+} + 2e^- \longrightarrow Cu(S)$

  Reaction occurring at the anode: $Cu(S) \longrightarrow Cu^{2+} + 2e^-$

Advantages –

i) High Purity: Electrolytic refining allows for the production of extremely pure metals, often exceeding 99.9% purity.

ii) Selective Removal: This process selectively removes impurities, leaving behind a highly pure metal deposit.

iii) Scalability: Electrolytic refining can be scaled up for commercial production, making it suitable for large-scale metal purification.

Limitations-

i) Energy Consumption: Electrolytic refining requires a significant amount of electrical energy to drive the process, making it energy-intensive.

Cost: The equipment and infrastructure required for electrolytic refining can be expensive, especially for small-scale operations

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CORROSION

Introduction

Corrosion - Any process of destruction of a metal due to chemical or electrochemical reaction with the surrounding medium normally starting at the surface is called as corrosion.

Causes of corrosion: -

1. All metals except (Pt, Au, Ag) occurs in a combined state in nature. During extraction of metal from its ore, they are reduced to metallic compound using energy. Hence an isolated metal will be present in its excited unstable state. However, all the metals have natural tendency to revert back to their combined state for its stability. As a result, metal exposed to environmental factors or chemicals get converted in to its original combined state which is called corrosion.so Corrosion is the reverse of extraction of metals and vice versa.

$$M + O \underset{\text{Extraction}}{\overset{\text{Corrosion}}{\rightleftharpoons}} MO + \text{Energy}$$

The corrosion process is a reversible process and is measured in units such as milli inches / year or millimeter/year.
2. Reactivity of metal-Highly reactive metals corrode faster.
3. Homogeneity of the metallic structure
4. Nature of the corrosive environment
5. Presence of air, moisture, gases like SO_2 and CO_2 near metal.
6. Presence of impurity-Presence of salts like NaCl etc.of electrolytes.
7. Characteristics of water in contact.(Like alkalinity of it / presence of ions / hardness / pH etc.
8. Bacteria in surrounding (It's presence increase O_2 concentration)

Types of corrosion

(i) Dry(Chemical corrosion)
(ii) Wet(Electrochemical corrosion)

Dry corrosion

Any process of destruction of a metal due to chemical reaction with the surrounding medium in absence of moisture or a conducting medium normally starting at the surface is called as corrosion.
It is of three types:-
(a) Oxidation corrosion
(b) Corrosion due to other gases
(c) Liquid metal corrosion

(a)Oxidation corrosion

This takes place when the metal is attacked by oxygen gas in absence of moisture. At ordinary temperatures metals are slightly attacked with the exception of alkali and alkaline earth metals. At high temperatures almost all the metals are oxidized with the exception of noble metals.

Mechanism of oxidation corrosion

Suppose M is a bivalent metal. The reactions when the metal is attacked by oxygen are:

$$M \rightarrow M^{2+} + 2e$$
$$\frac{1}{2} O_2 + 2e \rightarrow O^{2-}$$

Overall reaction $M + \frac{1}{2} O_2 \rightarrow M^{2+} + O^{2-}$
or $M^{2+} + O^{2-} \rightarrow MO$

Oxidation first occurs at the surface of the metal and the metal oxide so formed creates a barrier between the metal and oxygen. For oxidation to continue either the metal ions must diffuse

out or oxygen should penetrate inside. Generally the metal ions diffuse because of their smaller size. The nature of the oxide film formed decides whether oxidation will continue further.

There are four types of oxide films, namely (i) stable (ii) unstable (iii) Volatile (iv) Porous

i) **Stable film:** It is fine grained structure and sticks firmly to the surface. Hence it is nonporous or impervious in nature and further oxidation is prevented. Examples are: Al,Sn,Pb,Cu,Cr,Ni,Ti etc.

ii) **Unstable film:** The oxide layer formed decomposes back into the metal and oxygen.

Metal Oxide = Metal + Oxygen

Thus corrosion is not possible in such cases. Examples are: Ag, Au, and Pt.

(iii) **Volatile film :** The oxide layer volatilizes as soon as it is formed, thereby leaving the underlying metal exposed for further attack. This causes rapid and continuous corrosion. In this way very intensive corrosion takes place. Examples are: Molybdenum.

(iv) Porous film: This type of film is not fine grained (porous) so access of oxygen to the metal is not prevented and corrosion continues till the whole metal is completely converted to its oxide. Examples: all alkali and alkaline earth metals.

Corrosion due to other gases

Gases like CO_2, SO_2, Cl_2, H_2S, F_2 etc present in atmosphere generally in industrial areas have different affinities for different metals and just like oxidation corrosion various types of carbonate, sulphate, chloride, sulfide and fluoride films are formed. These films can be broadly classified as (i) protective and (ii) non- protective. Protective film's example is AgCl while nonprotective film's example is $SnCl_4$. Nonprotective or porous film is highly detrimental(dangerous) for the life of the metal.

Pilling-Bedworth rule

- It states that "an oxide layer is protective or nonporous , if the volume of the oxide is at least as great as the volume of the metal from which it is formed".
- Metals like Al, Sn etc. form oxide films whose volume is greater than the volume of the metal attacked. As a result highly adherent, non-porous film is formed and further corrosion is prevented.
- On the other hand "if the volume of the oxide film is less than the volume of the metal from which it is formed the oxide layer is porous and hence nonprotective".
- For ex. Alkali and alkaline earth metals undergo corrosion very fast because they form oxide films whose volume is less than the volume of the metal attacked resulting in cracks and pores

due to the stress formed. Oxygen enters through these cracks and corrosion continues non-stop till the whole metal is converted into its oxide.

$$PBR = \frac{Volume\ of\ Metal\ oxide\ film}{Volume\ of\ metal\ consumed\ in\ the\ formation\ of\ MO}$$

Interpretation of the PBR:

- **PBR < 1**: The oxide layer is too thin and does not cover the metal surface fully, leading to poor protection against further oxidation.
- **PBR ≈ 1 to 2**: The oxide layer is generally protective, as it adheres well and prevents further oxidation.
- **PBR > 2**: The oxide layer is too thick and tends to crack or spall off, which also exposes the metal to further oxidation.

Electrochemical corrosion/ wet corrosion

Any process of destruction of a metal due to chemical or electrochemical reaction with the surrounding medium normally starting at the surface in presence of conducting medium is called as wet corrosion. It takes place through:

(a) Electrochemical action
(b) Galvanic action or
(c) Concentration cell action

Electrochemical corrosion is characterized by –

1. The presence of conducting medium.
2. The formation of anodic and cathodic areas or parts in contact with each other.
3. Corrosion of anodic areas only.
4. Formation of corrosion product somewhere between anodic and cathodic areas

MECHANISM OF ELECTROCHEMICAL CORROSION

Electrochemical corrosion involves flow of electron current between the anodic and cathodic areas.

Reaction on anode (Oxidation)

The anodic reaction involves-

Dissolution of metal into metal ion with the liberation of free electrons.

$M\ (S) \longrightarrow M^{n+}\ (aq.) + ne^-$

$Fe\ (S) \longrightarrow Fe^{2+}\ (aq.) + 2e^-$

Reaction on cathode (Reduction)

Cathodic reaction involves the gain of electrons either by –

i) Evolution or liberation of hydrogen

ii) Absorption of Oxygen

i) Evolution or liberation of hydrogen

Evolution or liberation of hydrogen- type corrosion occurs in acidic environments.
In acidic medium H^+ ions gain electrons and there is liberation of hydrogen gas.

At Cathode $2\ H^+ + 2e^- \longrightarrow H_2(g)$

Overall reaction $Fe\ (S) + 2\ H^+ \longrightarrow Fe^{2+}\ (aq.) + H_2(g) \uparrow$

Thus, this type of corrosion causes- "displacement of hydrogen ions from the acidic solution by metal ions.

All metals above hydrogen in electrochemical series have a tendency to get dissolve in acidic solution with simultaneous evolution of hydrogen.

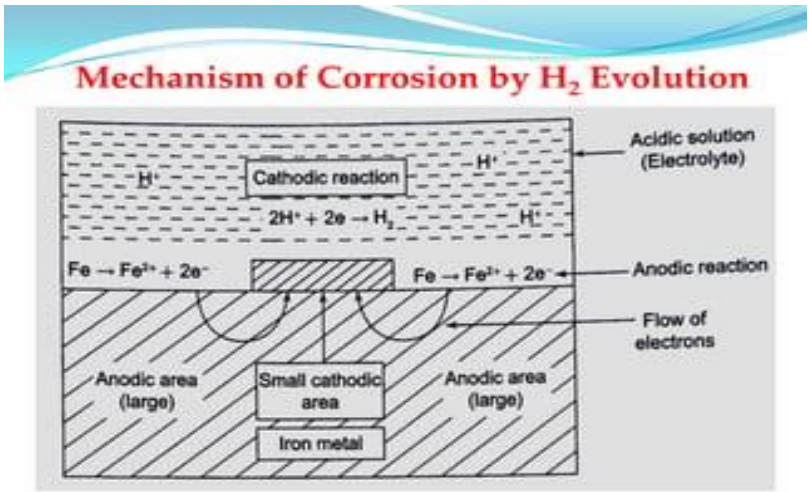


Fig: Corrosion by evolution of hydrogen

ii) Absorption of Oxygen

This type of reaction occurs in Neutral or slightly alkaline medium.

Example rusting of iron in neutral aqueous solution of electrolytes (like NaCl) in the presence of atmospheric oxygen.

a) **Reaction at anode** : Iron dissolves as ferrous ions with liberation of electrons

$Fe\ (S) \longrightarrow Fe^{2+}\ (aq.) + 2e^-$

The liberated electrons flows from anodic to cathodic areas

a) Reaction at cathode :

$\frac{1}{2}\ O_2 + H_2O + 2e^- \longrightarrow 2\ OH^-$

b) Overall reaction

$$Fe\ (S) + \tfrac{1}{2}\ O_2 + H_2O \longrightarrow Fe^{2+} + 2\ OH^-$$

The Fe^{2+} ions from anode and OH^- ions at cathode diffuse towards each other through conducting medium and form ferrous hydroxide.

$$Fe^{2+}(aq.) + 2\ OH^- \longrightarrow Fe(OH)_2 \downarrow$$

Ferrous hydroxide

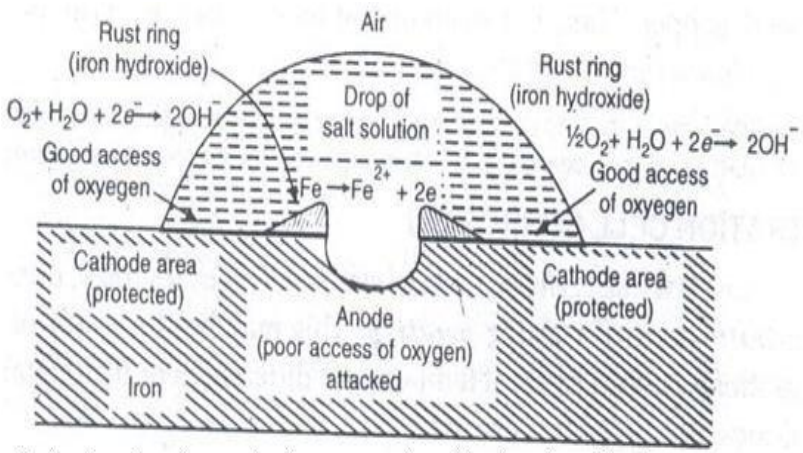
i) If enough oxygen is present ferrous hydroxide is easily oxidized to ferric hydroxide.

$$4Fe(OH)_2 + O_2 + 2\ H_2O \longrightarrow 4\ Fe\ (OH)_3$$

Ferric hydroxide.

This product is called yellow rust which is actually corresponds to $Fe_2O_3.H_2O$.

ii) If supply of oxygen is limited the corrosion product may be even black anhydrous Magnetite Fe_3O_4.



Mechanism of wet corrosion by oxygen adsorption (rusting of iron)

Concentration cell action

- The electrochemical theory could successfully explain the corrosion of metals in contact with a more noble metal or alloy.
- It could also explain why uneven or scratched surfaces undergo corrosion. However this theory could not explain why corrosion often occurs at places which are not easily accessible to oxygen.
- For this a modified electrochemical theory is suggested by scientists. The theory stated that corrosion takes place in many cases due to different amounts of oxygen or "differential aeration".

- Anodic and cathodic areas may be formed even in a perfectly polished, homogeneous and pure metal due to different amounts of oxygen reaching different parts of the metal forming Oxygen concentration cell.
- In case of Oxygen concentration cell the poorly oxygenated part becomes anodic and naturally more oxygenated part becomes cathodic and is protected. A flow of electrons from anode to cathode takes place called as "differential current".

Examples of differential aeration or concentration cell action

(1) If an iron pipe is incompletely covered by an oxide surface layer then in presence of moisture, a concentration cell is set up & the metal beneath the oxide layer becomes anodic and is corroded.

(2) Corrosion is accelerated under accumulation of dirt, sand, scale or other contaminants. This is because accumulation of various impurities restricts the access of oxygen to those parts & establishes an anode to promote still greater accumulation. This results in localized corrosion, due to non- uniform corrosion. (pitting)

(3) Metals exposed to aqueous media corrode under blocks of wood or pieces of glass which screen that metal from oxygen.